(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :14/10/2025

(21) Application No.202541099319 A

(43) Publication Date : 28/11/2025

(54) Title of the invention : Smart Traffic Management System for Urban Congestion

(51) International classification	:G08G0001010000, G08G0001080000, G08G0001042000, H04L0067120000, H04W0004700000	(71)**Name of Applicant :** **1)MURUGALAKSHMI k** Address of Applicant :Assistant Professor, Department of Artificial Intelligence and Data Science, VSB College of Engineering Technical Campus, Kinathukadavu. Tamil Nadu India **2)Dr. R. Murugadoss**
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(87) International Publication No	: NA	**4)Ms. Hemasri S**
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(57) Abstract :
Abstract Traditional traffic signal systems rely heavily on fixed-time control mechanisms that operate on predetermined schedules, regardless of actual traffic conditions. These systems typically use basic sensors like inductive loops or infrared detectors to detect vehicle presence, but their integration is limited and lacks real-time adaptability. Traffic monitoring is often conducted manually or through periodic CCTV footage reviews, which do not offer dynamic control or immediate responsiveness. Public access to traffic updates is generally provided through radio broadcasts or static online maps, offering limited support for real-time route planning. While these conventional methods have served urban infrastructure for decades, they fall short in addressing modern traffic challenges such as congestion, emergency routing, and dynamic traffic flow optimization. The absence of centralized control and intelligent decision-making leads to inefficiencies, longer commute times, and increased fuel consumption. Without leveraging advanced technologies like artificial intelligence, IoT, or cloud-based analytics, these systems remain reactive rather than proactive, making them outdated in the face of growing urban mobility demands.

No. of Pages : 11 No. of Claims : 10